

PART A — Reference sheets

Keep these pages. Do not write on them and do not hand them in.

Part A holds the notation, the formula sheet and the complete R session that every question reads from. The questions are in **Part B**, which starts on a separate sheet, carries the name line, and is the only part you hand in: **25 questions in 40 minutes**.

Notation used on this paper. (The formulas behind each symbol are on the formula sheet below, F1–F11.)

- **X** is the $n \times p$ **data matrix**: n rows (units) and p columns (variables); x_{ik} is the value of variable k on unit i . In R it is `X`.
- $\bar{\mathbf{x}} = (\bar{x}_1, \dots, \bar{x}_p)^\top$ is the **sample mean vector**, one column mean per variable (F1). In R: `xbar <- colMeans(X)`.
- s_k is the **sample standard deviation** of variable k (F2). In R: `apply(X, 2, sd)`.
- **S** is the $p \times p$ **sample covariance matrix** with entries s_{jk} (F3); its diagonal holds the variances $s_{kk} = s_k^2$. In R: `S <- cov(X)`.
- **R** is the $p \times p$ **sample correlation matrix** with entries r_{jk} (F4). In R: `R <- cor(X)`.
- **Z** is the **standardised** matrix with entries z_{ik} (F7): every column has mean 0 and standard deviation 1. In R: `Z <- scale(X)`.
- $d(\mathbf{x}, \mathbf{y})$ is the **Euclidean distance** between two rows (F8). `dist(X)` returns it for every pair of rows, printed as a lower triangle: the entry in row i , column j is $d(\text{row } i, \text{row } j)$.
- d_i^2 is the **squared Mahalanobis distance** of unit i from the centre $\bar{\mathbf{x}}$ (F10, Lecture 5). In R: `d2 <- mahalanobis(X, xbar, S)`, which returns one number per row.

Formula sheet (Lectures 1–5). Sums over i run over the n rows, sums over k over the p columns; $j, k \in \{1, \dots, p\}$ index variables.

F1	Sample mean of variable k	$\bar{x}_k = \sum_i x_{ik} / n$
F2	Sample variance; standard deviation	$s_k^2 = \sum_i (x_{ik} - \bar{x}_k)^2 / (n - 1); \quad s_k = \sqrt{s_k^2}$
F3	Sample covariance ($s_{kk} = s_k^2$)	$s_{jk} = \sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k) / (n - 1)$
F4	Sample correlation	$r_{jk} = s_{jk} / (s_j s_k) = s_{jk} / \sqrt{s_{jj} s_{kk}}$
F5	Textbook (divisor n) covariance matrix	$\mathbf{S}_n = \frac{n-1}{n} \mathbf{S}$
F6	Generalized variance; total variance	$ \mathbf{S} $ (determinant); $\text{tr}(\mathbf{S}) = \sum_k s_{kk}$
F7	Standardised score	$z_{ik} = (x_{ik} - \bar{x}_k) / s_k$
F8	Euclidean; city-block; Chebyshev	$\sqrt{\sum_k (x_k - y_k)^2}; \quad \sum_k x_k - y_k ; \quad \max_k x_k - y_k $
F9	Euclidean distance on standardised columns	$\sqrt{\sum_k ((x_k - y_k) / s_k)^2}$
F10	Squared Mahalanobis distance from $\bar{\mathbf{x}}$	$d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$
F11	Squared Mahalanobis distance, two rows	$d^2(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i - \mathbf{x}_j)^\top \mathbf{S}^{-1} (\mathbf{x}_i - \mathbf{x}_j)$

The R session. Every question refers to the four blocks below. The data are `aq`, the **111 complete days** of R's `airquality` record at a New York monitoring station: `Ozone` (parts per billion), `Solar.R`

(langleys), Wind (mph) and Temp (degrees Fahrenheit). Lines beginning > are what was typed; the rest is what R printed. *Two functions you have not had to type:* `t()` is R's transpose — it swaps the rows and the columns of a table; `rowMeans()` is the row twin of `colMeans()` — it averages *across* each row instead of down each column.

Block 1 — the data. (`complete.cases()` keeps the rows with no missing value.)

```
> aq <- airquality[complete.cases(airquality), ] # keep complete rows
> aq <- aq[, c("Ozone", "Solar.R", "Wind", "Temp")]
> dim(aq)
[1] 111 4
> str(aq)
'data.frame': 111 obs. of 4 variables:
 $ Ozone : int 41 36 12 18 23 19 8 16 11 14 ...
 $ Solar.R: int 190 118 149 313 299 99 19 256 290 274 ...
 $ Wind : num 7.4 8 12.6 11.5 8.6 13.8 20.1 9.7 9.2 10.9 ...
 $ Temp : int 67 72 74 62 65 59 61 69 66 68 ...
```

Block 2 — centre, spread, association.

```
> X <- as.matrix(aq)
> xbar <- colMeans(X)
> round(xbar, 2)
  Ozone Solar.R   Wind   Temp
 42.10 184.80   9.94  77.79
> S <- cov(X)
> round(S, 2)
      Ozone Solar.R   Wind   Temp
Ozone 1107.29 1056.58 -72.51 221.52
Solar.R 1056.58 8308.74 -41.24 255.47
Wind -72.51 -41.24 12.66 -16.86
Temp 221.52 255.47 -16.86 90.82
> R <- cor(X)
> round(R, 3)
      Ozone Solar.R   Wind   Temp
Ozone 1.000 0.348 -0.612 0.699
Solar.R 0.348 1.000 -0.127 0.294
Wind -0.612 -0.127 1.000 -0.497
Temp 0.699 0.294 -0.497 1.000
> det(R)
[1] 0.2708813
> round(apply(X, 2, sd), 2)
  Ozone Solar.R   Wind   Temp
33.28  91.15   3.56   9.53
```

Block 3 — how unusual is each day? Squared Mahalanobis distance from $\bar{x}$, the five largest, and the columns' ranges.

```
> d2 <- mahalanobis(X, xbar, S)
> round(sort(d2, decreasing = TRUE)[1:5], 3)
      117      48      9      62      18
25.077 14.499 13.521 11.119  9.801
> X[c("117", "48", "9"), ]
      Ozone Solar.R Wind Temp
117    168     238   3.4   81
48     37     284  20.7   72
9       8      19  20.1   61
> apply(X, 2, range)
      Ozone Solar.R Wind Temp
[1,]      1       7   2.3   57
[2,]    168    334  20.7   97
```

Block 4 — Euclidean distances between four of those days, on the raw columns and then on the standardised columns.

```
> four <- X[c("117", "48", "9", "62"), ]
> round(dist(four), 1)
      117      48      9
48 140.2
9  272.5 266.8
62  45.4 101.2 281.8
> Z <- scale(X)
> round(dist(Z[c("117", "48", "9", "62"), ]), 2)
      117      48      9
48  6.35
9   7.44  3.25
62  1.11  5.66  6.94
```

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Qatar University — Department of Mathematics and Statistics

STAT 481 — Multivariate Analysis — Sample Quiz 1: Data, Summary, Display,
Distance — SAMPLE

Lectures L1–L5 — Section L01 (Version S)

Fall 2026 — corrected in class Tue, Sep 8, 2026

NAME: _____ STUDENT ID: _____

PART B — Questions

Total Questions: 25 Each Worth: 1 point Time allowed: 40 minutes

Instructions: Fill the bubble of **every** statement you judge correct: a question may have one correct answer or more, and **deciding how many is part of the question**. No calculators needed. **You are not running R during this quiz:** every piece of output you need is printed in **Part A** (notation, formula sheet, four R blocks), and the questions refer back to it. **Hand in Part B only.**

Filling more bubbles than a question has correct answers voids that question.

Mark your answers as shown: ☐ ☐ ☒ ☐ (completely fill the selected bubble)

Q.1 `cor(aq)` returns a 4×4 matrix.

☐ True

☐ False

Q.2 Sorting the 111 days by `Temp` before calling `cor()` would change at least one entry of **R** in Block 2.

☐ True

☐ False

Q.3 Re-expressing `Temp` in degrees Celsius leaves every entry of **R** in Block 2 unchanged.

☐ True

☐ False

Q.4 `cov(t(aq))` stops with an error.

☐ True

☐ False

Q.5 From Block 1, the n and p of the data matrix used on this paper are

☐ $n = 153$ and $p = 6$

☐ $n = 111$ and $p = 6$

☐ $n = 4$ and $p = 111$

☐ $n = 111$ and $p = 4$

Q.6 The code below computes the covariance between `Ozone` and `Temp` from its definition, without calling `cov()`. Line 4 is incomplete.

```
1 n <- nrow(X)
2 oz <- X[, "Ozone"] - mean(X[, "Ozone"])
3 te <- X[, "Temp"] - mean(X[, "Temp"])
4 s_OT <-
```

Which completion of line 4 makes `s_OT` equal to the entry 221.52 printed in Block 2?

☐ `sum(oz * te) / n`

☐ `sum(oz) * sum(te) / (n - 1)`

☐ `sum(oz * te) / (n - 1)`

☐ `sum(oz * te)`

- Q.7** Two analysts rank the same 111 days by how far each is from $\bar{\mathbf{x}}$. One uses the Euclidean distance on the raw columns; the other uses d^2 . Use the standard deviations printed in Block 2. Which statements hold?
- ☐ The Euclidean ranking is very nearly a ranking on `Solar.R` alone.
 - ☐ The two rankings must agree, because both start from the same $\bar{\mathbf{x}}$.
 - ☐ Multiplying every `Wind` value by 100 changes the Euclidean ranking but not the d^2 ranking.
 - ☐ Both rankings are free of the units in which the columns were recorded.
- Q.8** Block 2 prints `det(R)` as 0.2708813. Which statement is supported?
- ☐ About 27% of the variability in the data has been accounted for.
 - ☐ The four variables carry appreciably less than four variables' worth of separate information.
 - ☐ The four variables are mutually uncorrelated.
 - ☐ The generalized variance of the raw data is 0.2708813.
- Q.9** The strongest association in the correlation matrix of Block 2 is between
- ☐ `Ozone` and `Temp`
 - ☐ `Ozone` and `Solar.R`
 - ☐ `Ozone` and `Wind`
 - ☐ `Wind` and `Temp`
- Q.10** Which single expression returns the four sample variances of `aq`, and nothing else?
- ☐ `cov(aq)`
 - ☐ `diag(cov(aq))`
 - ☐ `diag(cor(aq))`
 - ☐ `apply(aq, 1, var)`
- Q.11** A colleague reads the correlation matrix in Block 2 and asks what it says about the four measurements. Which statements could you defend?
- ☐ Wind speed has essentially no linear relation with any of the other three variables.
 - ☐ Hot days tend to be high-ozone days in this record.
 - ☐ The table establishes that high temperature drives ozone formation.
 - ☐ Solar radiation is the least connected of the four to the rest of the table.
- Q.12** The code below was written to reproduce the `d2` of Block 3 — the squared Mahalanobis distance of every day from $\bar{\mathbf{x}}$. It does not do that.

```

1 X <- as.matrix(aq)
2 xb <- rowMeans(X)
3 S <- cov(X)
4 d2 <- mahalanobis(X, xb, S)

```

Which line must be corrected?

- ☐ line 1
- ☐ line 2
- ☐ line 3
- ☐ line 4

Q.13 The line below is meant to turn the covariance 221.52 of Block 2 into the correlation 0.699 of the same block, using `S`. It is incomplete.

```
1 r_OT <- S["Ozone", "Temp"] /
```

Which completion is correct?

- ☐ `sqrt(S["Ozone", "Ozone"] * S["Temp", "Temp"])`
- ☐ `(S["Ozone", "Ozone"] * S["Temp", "Temp"])`
- ☐ `(sqrt(S["Ozone", "Ozone"]) + sqrt(S["Temp", "Temp"]))`
- ☐ `det(S)`

Q.14 Which statements follow from Block 3?

- ☐ Day 117 is the most unusual of the 111 days on these four measurements.
- ☐ Day 117's 25.077 against day 18's 9.801 makes day 117 about 2.5 times as far from $\bar{x}$ as day 18.
- ☐ The output identifies `Wind` and `Temp` as the two variables responsible for day 9.
- ☐ Day 9 is unusual although none of its four values is extreme in its own column.

Q.15 A colleague asks: “*which of these four measurements are telling me the same thing?*” Which single line best answers that question?

- ☐ `stars(aq)`
- ☐ `faces(aq)`
- ☐ `ggpairs(aq)`
- ☐ `plot(aq$Ozone, aq$Temp)`

Q.16 Read the first table of Block 4, `round(dist(four), 1)`. Which statements hold?

- ☐ The upper half of the table is blank because those distances are undefined.
- ☐ Of the four days, 117 and 62 are the closest pair.
- ☐ Day 9 is farther from day 117 than from day 48.
- ☐ The entry 140.2 is measured in parts per billion.

Q.17 In Block 4 the four days are compared again after `scale()`. After standardising, the day closest to day 48 is

- ☐ day 117
- ☐ day 9
- ☐ day 62
- ☐ cannot be told from the table

Q.18 A colleague wants the distance between every pair of days on the four columns of `aq`, and shows you the standard deviations in Block 2. By the workflow of Lecture 5, the call to make is

- ☐ `dist(X)`
- ☐ `scale(dist(X))`
- ☐ `dist(X, method = "maximum")`
- ☐ `dist(scale(X))`

Q.19 `dim(scale(X))` prints

- ☐ `[1] 4 4`
- ☐ `[1] 111 1`
- ☐ `[1] 4 111`
- ☐ `[1] 111 4`

Q.20 Use the means in Block 2 and the rows and ranges in Block 3. Which statements hold?

- ☐ Day 117's `Ozone` value is the largest `Ozone` of the 111 days.
- ☐ Every day's `Temp` lies within 20 degrees of the mean `Temp`.
- ☐ The mean `Solar.R` is closer to the largest `Solar.R` than to the smallest.
- ☐ $\bar{x}$ is the day with the smallest d^2 .

Q.21 Block 3 prints the five largest values of `d2`. What does it tell you about the *other* 106 days?

- ☐ Each of them has d^2 below 1.
- ☐ Nothing — their values cannot be bounded from a sorted top five.
- ☐ Each of them has d^2 at most 9.801.
- ☐ They are the 106 days with the smallest `Solar.R` values.

Q.22 Which single line returns the raw Euclidean distance 140.2 between day 117 and day 48 printed in Block 4, and nothing else?

- ☐ `dist(X)[ "117", "48" ]`
- ☐ `sum(abs(X[ "117", ] - X[ "48", ] ))`
- ☐ `mahalanobis(X[ "117", ], X[ "48", ], S)`
- ☐ `dist(X[c( "117", "48" ), ])`

Q.23 The code below was written to reproduce the *second* table of Block 4. It runs, but prints different numbers.

```
1 X      <- as.matrix(aq)
2 four   <- X[c( "117", "48", "9", "62" ), ]
3 Z      <- scale(four)
4 round(dist(Z), 2)
```

Which line must be corrected?

- ☐ line 1
- ☐ line 2
- ☐ line 3
- ☐ line 4

Q.24 `Z <- scale(X)` as in Block 4. Which statements hold?

- ☐ `colMeans(Z)` is 0 0 0 0 up to rounding.
- ☐ `cov(Z)` is equal to the correlation matrix `R` of Block 2.
- ☐ `cor(Z)` is the 4×4 identity matrix.
- ☐ `max(Z)` is at most 1: standardising maps every column into $[-1, 1]$.

Q.25 By the decision table of Lecture 5, `dist(X)` on the raw columns — *no* standardising — is the right call for which of these tables?

- ☐ Four exam marks, each out of 100, for 111 students.
- ☐ The four columns of `aq` exactly as recorded.
- ☐ Any table in which n is much larger than p .
- ☐ None — every call to `dist()` must be preceded by `scale()`.